Red-Blue Hamiltonian Tomography via the Operator Lightcone Wavemap

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Abstract

We introduce a model-free method for Hamiltonian tomography based on the operator spreading wavefront. Given two tensor network simulations of the same quantum lattice (one with the nominal Hamiltonian and one with one or more perturbed bonds), the signed cross-entropy of the non-identity Pauli weight field $\mathcal{L}(v,t) = -\ln(n_{\text{pert}}/n_{\text{base}})$ identifies each perturbed bond by its causal signature: a red-blue sign flip at the two endpoints, visible from the first wavefront crossing. Combining measurements from p strategically placed observation centers via an ensemble log-likelihood sum, a single broken bond is identified with an 82-fold margin over the next candidate on a 7×7 lattice. Extending to realistic coupling degradation ($\delta J = -0.20$) and up to 8 simultaneous faults, the integrated log-ratio $\Lambda(v) = \ln(\mathcal{B}(v)/\mathcal{N}(v))$ achieves a per-bond detection rate above 90% in a single seismic pass using only 6 simulations. An iterative calibration loop reduces multi-bond detection to a sequence of single-bond problems, each solved at 100% accuracy. The method requires no model assumptions and no knowledge of the perturbation type or magnitude; the detection is deliberately blind.

1 Introduction

Identifying miscalibrated or anomalous couplings in a quantum processor is a central challenge in quantum hardware characterization. Standard approaches require explicit Hamiltonian learning: fitting a parameterized coupling model to measurement data. This scales poorly with system size and depends on model assumptions about the noise and interaction structure.

We propose a different approach based on a clear separation of roles. The *simulator* knows the nominal Hamiltonian and produces two Pauli weight fields: $n_{\text{base}}(v,t)$ from the nominal evolution, and $n_{\text{pert}}(v,t)$ from an evolution where one bond has been silently modified. The *detection algorithm* receives only these two arrays of numbers and must identify the modified bond from their ratio alone, with no knowledge of the Hamiltonian, the perturbation type, or its magnitude. The detection is deliberately blind.

A perturbed bond creates a causal signature in the wavefront: a sign flip in the cross-entropy field $\mathcal{L}(v) = -\ln(n_{\text{pert}}/n_{\text{base}})$ between its two endpoints that propagates no further. The broken bond is exactly where the wavefront sign changes, identifiable from the observables alone.

The method is inspired by seismic tomography: place sensors (observation centers) at strategic positions, emit a wave (initialize a local operator), and read the arrival pattern. The perturbed bond is the anomaly in the medium.

The red-blue color scheme is a deliberate reference to the red-blue pebble game of Hong and Kung [2], which characterizes I/O complexity as a bottleneck in computational graphs. Here the roles are analogous: red sites (upstream, wave blocked) and blue sites (downstream, wave starved)

identify the bond that limits operator propagation, exactly as red and blue pebbles identify the I/O bottleneck in a computation.

2 Setup

We simulate the kicked isotropic Heisenberg model on a 7×7 rectangular lattice (49 sites, 84 bonds) using a GPU-accelerated belief propagation tensor network simulator with Trotter step $\varepsilon = 0.10$, bond dimension $\chi = 50$, and truncation cutoff $\delta = 10^{-3}$. The non-identity Pauli weight at site v and cycle t is:

$$n(v,t) = \sum_{P:\, P_v \neq I} c_P(t)^2 \tag{1}$$

where $O_c(t) = \sum_P c_P(t) P$ is the evolved operator initialized at center c. By normalization $\sum_P c_P^2 = 1$, $n(v,t) \in [0,1]$ is a probability.

To simulate miscalibrated bonds, we weaken their couplings by δJ, with each bond potentially taking a different value. In the experiments below we use $\delta J = -0.20$ ($J = 0.80$, a 20% degradation) as a representative choice; the detection method does not depend on knowing δJ in advance and works for any coupling deviation. The Pauli transfer matrix entries scale as $\cos(J\varepsilon)$, attenuating operator propagation without blocking it entirely. For the multi-bond experiments in Section 10, k bonds are selected uniformly at random and perturbed simultaneously.

We place $p = 6$ observation centers using a greedy geodesic bond cover: the minimum set of sites such that every bond is within $d_{\max} = 2$ hops of at least one center. For the 7×7 lattice this gives centers at sites $\{6, 10, 25, 30, 32, 34\}$, covering all 84 bonds.

3 Signal Vocabulary and Measurement Definitions

All definitions are expressed in terms of two observables per site per cycle: $n_{\text{base}}(v,t)$ and $n_{\text{pert}}(v,t)$, the non-identity Pauli weights under the nominal and perturbed Hamiltonian respectively. The BFS frontier crossing time $t_0(v) = \min\{t : n_{\text{base}}(v,t) > \varepsilon_{\text{th}}\}$ defines a site-specific reference time; it marks the physical arrival of the wavefront at v, set by the Lieb-Robinson lightcone, not by the numerical cutoff.

At each site v in the 3-cycle window $[t_0(v) \pm 1]$ we observe one of three frontier patterns. A site is **White** [W] when $n_{\text{base}}(v, t_0) \leq \varepsilon_{\text{th}}$: the baseline wave has not yet arrived and no information is available. A site is **Blue** [B] when $n_{\text{pert}}(v, t_0) > n_{\text{base}}(v, t_0) > \varepsilon_{\text{th}}$: the perturbed wave accumulates at v because drainage is blocked by a broken bond downstream; the site is a *wall face*. A site is **Red** [R] when $n_{\text{pert}}(v, t_0) < n_{\text{base}}(v, t_0)$ with $n_{\text{base}}(v, t_0) > \varepsilon_{\text{th}}$: the perturbed wave is attenuated because a broken bond on the path from center blocked transmission; the site is in the *shadow*.

For a bond (u, v) with u on the center-side and v on the shadow-side, these patterns combine into three observable signatures. When both endpoints are white [W, W, W] the bond has not yet been reached or is healthy. When the wall face is detected but the shadow is not yet visible [W, B, W], one more cycle is needed. The **broken bond signature** [W, B, R] is the primary target: u appears blue at cycle t_0 and v appears red at cycle $t_0 + 1$, identifying the bond between them as anomalous. This pattern rests on a single assumption: both waves reach u at the same frontier time t_0, so any amplitude difference at v is attributable to the bond, not to a timing offset. The temporal order (blue before red by one cycle) is then universal, following from the BFS causal structure: u is reached before v, so the wall accumulates before the shadow appears.

The signed cross-entropy evaluated in the frontier window is:

$$\mathcal{L}(v_0,\,[t_0 \pm 1]) = -\ln \frac{n_{\text{pert}}(v_0,\,t_0)}{n_{\text{base}}(v_0,\,t_0)} \tag{2}$$

This is a *cross-correlation fixed in space*: both waves evaluated at the same site v_0 at the same frontier moment t_0, with the window $[t_0 \pm 1]$ providing context (white before, red after). Positive $\mathcal{L}$ identifies red (shadow), negative identifies blue (wall face). The broken bond is the edge between consecutive blue and red sites in the BFS ordering.

A complementary observable measures the wave *through* a bond rather than *at* a site. Consider the bond (v_0, u_0) in the BFS path from the center. The baseline reaches v_0 at frontier time t_0 and u_0 at $t_1 = t_0 + 1$. The *baseline transfer ratio* is:

$$T_{\text{base}}(v_0 \to u_0) = \frac{n_{\text{base}}(u_0,\,t_1)}{n_{\text{base}}(v_0,\,t_0)} \tag{3}$$

This measures how much of the wave at v_0 propagates through the bond to u_0 in one BFS step, directly encoding the gate's coupling strength J. For the perturbed run (with potentially delayed frontier times t_0^p, t_1^p):

$$T_{\text{pert}}(v_0 \to u_0) = \frac{n_{\text{pert}}(u_0,\,t_1^p)}{n_{\text{pert}}(v_0,\,t_0^p)} \tag{4}$$

The ratio $T_{\text{base}}/T_{\text{pert}} > 1$ signals a broken bond: less weight crosses the bond in the perturbed run.

Two important caveats apply. First, there is a path distortion. The perturbed wave arriving at v_0 has already traversed all upstream bonds. If any of those are also broken, $n_{\text{pert}}(v_0, t_0^p)$ is already weakened, not by bond (v_0, u_0) but by upstream damage. The ratio $T_{\text{base}}/T_{\text{pert}}$ then conflates the upstream distortion with the target bond signal. The ratio is not a pure measurement of bond (v_0, u_0). Second, there is a measurable Sensitivity. problem. The perturbed weight $n_{\text{pert}}(v_0, t_0^p)$ may be close to ε_{th}, the threshold, making the ratio numerically unreliable. The local cross-correlation $\mathcal{L}(v_0, [t_0 \pm 1])$ avoids both caveats by using the baseline as the reference; it measures the *relative* change at the frontier moment, not the absolute transfer.

4 The Cross-Entropy Wavemap Observable

The previous section introduced the signal vocabulary for a wave with a single origin: one center fires, the wavefront spreads, and the red-blue sign flip at the bond endpoints is the detection signal. Here we extend to the case where p centers fire independently, each producing its own wavemap. The key quantities are defined per-center, then combined into an ensemble.

For each center k and site v, we evaluate the signed cross-entropy in a 3-cycle window $[t_0 \pm 1]$ centered on the BFS frontier crossing $t_0(v) = \min\{t : n_{\text{base}}^k(v,t) > \varepsilon_{\text{th}}\}$:

$$\mathcal{L}_k(v,\,[t_0 \pm 1]) = -\ln \frac{n_{\text{pert}}^k(v,\,t_0(v))}{n_{\text{base}}^k(v,\,t_0(v))} \tag{5}$$

This is a *cross-correlation fixed in space*: both waves are evaluated at the same site v and the same frontier moment $t_0(v)$, defined by the baseline. The window $[t_0 - 1, t_0, t_0 + 1]$ captures three states — before arrival (WHITE), wall-face signal (BLUE or WHITE), and shadow signal (RED) — and $t_0(v)$ is the moment of maximum signal and minimum χ-saturation noise.

Consider a bond (u, v) with u closer to center k. At time t_0, the baseline wave has just crossed the bond and $n_{\text{base}}(u, t_0) \approx n_{\text{pert}}(u, t_0)$; both waves have reached u with equal amplitude. One cycle

later at $t_1 = t_0+1$, the weaker bond slows drainage from u: weight accumulates, $n_{\text{pert}}(u) > n_{\text{base}}(u)$, so $\mathcal{L}(u) < 0$ (blue, wall face). Simultaneously, in the baseline weight flows through bond (u, v) and arrives at v, while in the perturbed run less weight crosses: $n_{\text{pert}}(v, t_1) < n_{\text{base}}(v, t_1)$, so $\mathcal{L}(v) > 0$ (red, shadow). The temporal order is universal: blue at the wall face first, red at the shadow one cycle later. The broken bond sits between the blue and red sites, identified by the sign change (Figure 1). The contrast $|\mathcal{L}(u) - \mathcal{L}(v)|$ is maximum at t_1, providing the detection signal.

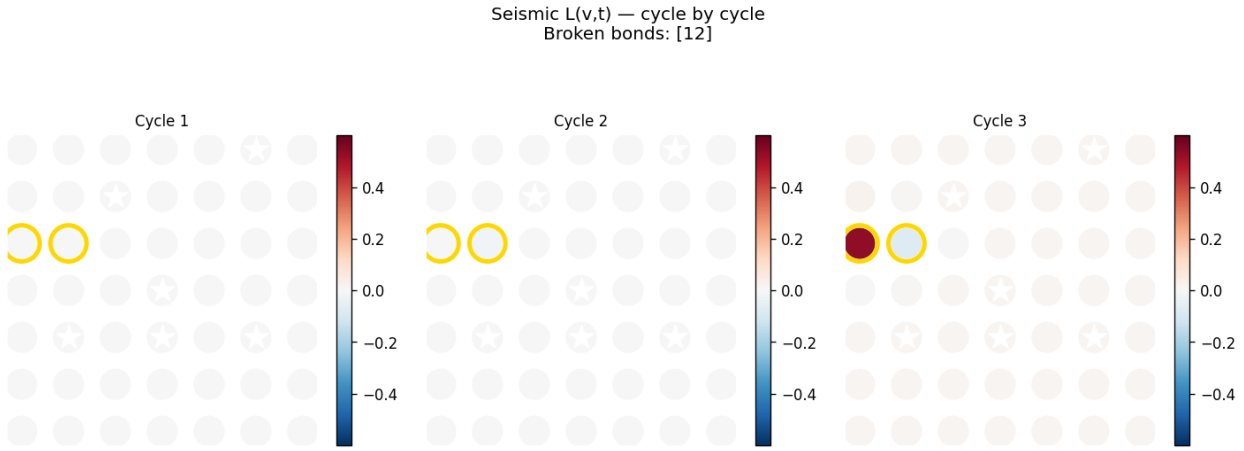


Figure 1: Signed cross-entropy $\mathcal{L}(v, t)$ for a single broken bond (bond 12, sites 15↔16, identity PTM), seismic initialization, rect 7×7, $\chi = 50$. **Cycle 1**: no signal. **Cycle 2**: site 16 red ($\mathcal{L} > 0$, shadow), site 15 blue ($\mathcal{L} < 0$, wall face): bond 12 identified by sign flip. **Cycle 3**: signal strengthens before χ saturation. Gold circles: bond endpoints. Stars: seismic centers.

Each center k is initialized and evolved *independently*: a separate baseline and perturbed simulation pair per center. The signed cross-entropy $\mathcal{L}_k(v)$ is computed for each center individually, then summed into an ensemble log-likelihood:

$$\mathcal{L}_{\text{ens}}(v) = \sum_{k=1}^{p} \mathcal{L}_k(v) \tag{6}$$

Centers on the same side of the broken bond reinforce each other; centers on opposite sides contribute opposite signs that partially cancel. Silent centers (wave never reaches site v) contribute zero and do not corrupt the sum.

A more robust alternative to the pointwise $\mathcal{L}(v, t_0)$ accumulates the signal over all T cycles:

$$\mathcal{B}(v) = \sum_{t=1}^{T} n_{\text{base}}(v, t), \qquad \mathcal{N}(v) = \sum_{t=1}^{T} n_{\text{pert}}(v, t), \qquad \Lambda(v) = \ln \frac{\mathcal{B}(v)}{\mathcal{N}(v)} \tag{7}$$

Positive Λ identifies red (shadow), negative identifies blue (wall face). The integral captures both the delay and the sustained amplitude difference: a broken bond delays the perturbed wave *and* reduces its steady-state weight, making $\Lambda(v)$ more robust than $\mathcal{L}(v, t_0)$ when the pointwise signal is weak. In the multi-bond experiments, $\Lambda(v)$ is the primary detection signal.

5 Bond Detection

For each bond b connecting sites v_a and v_b we define a *bond score*:

$$S(b) = \mathcal{L}_{\text{ens}}(v_a) \times \mathcal{L}_{\text{ens}}(v_b) \tag{8}$$

A negative product indicates a sign flip across the bond, the causal signature of the broken bond. Rank bonds by most negative $S(b)$.

Table 1 shows the individual and ensemble results for bond 12 (sites 15 $leftrightarrow$16, identity PTM perturbation, 7×7 lattice, $\chi = 50$, 5 cycles).

Center	Side	$\mathcal{L}(15)$	$\mathcal{L}(16)$	Result
10	upstream	+0.87	−0.02	✓CORRECT
30	upstream	+0.37	−0.04	✓CORRECT
25	downstream	+0.03	−0.04	× WRONG
32	downstream	+0.00	−0.03	× WRONG
6	too far	0	0	silent
34	too far	0	0	silent
Ensemble	—	+1.23	−0.13	✓CORRECT

Table 1: Signed cross-entropy at bond 12 endpoints (sites 15,16). Ensemble score: $S(12) = -0.164$, next bond $= -0.002$ (82× margin).

The ensemble identifies bond 12 with an 82-fold margin over the next candidate, despite only 2 of 6 individual centers giving the correct answer. The wrong individual answers: centers 25 and 32 both correctly identify site 16 as a bond endpoint; their error is only in declaring the wrong adjacent bond. The ensemble product cancels these partial errors.

6 Seismic Multi-Center Scanning

Rather than running each center independently, we initialize all p centers simultaneously in a single seismic scan: the initial operator is $O(0) = \sum_k Z_{c_k}$, firing all centers at once. At each site v, the first-arrival signal is dominated by the nearest center, automatically selecting the strongest directional component. The sign flip at the broken bond endpoints emerges without any explicit path tracing or per-center bookkeeping, and the cost drops from $2p$ simulations to just 2 (one baseline, one perturbed).

To achieve broad coverage with no geometric blind spots, we use three complementary center placements on the 7×7 lattice: corners and center ($\{1, 7, 25, 43, 49\}$), quadrant centers ($\{11, 13, 37, 39\}$), and mid-edge sites ($\{4, 22, 28, 46\}$). Each placement covers a different region of the grid; together the three scans ensure every bond is within $d_{\max} = 2$ hops of at least one center in at least one scan. The total cost is 6 simulations (3 baselines + 3 perturbed), run in parallel on 3 GPUs.

Bond scores are computed using the integrated log-ratio $\Lambda(v)$ across all three scans. For each bond b connecting sites v_a and v_b, the score is the maximum contrast $|\Lambda(v_a) - \Lambda(v_b)|$ over all scans. The top-k bonds by score are declared broken, where k is the number of expected faults. On 20 single-bond trials ($\delta J = -0.20$, two independent datasets), this procedure achieves **20/20** correct identifications. On 10 two-bond trials it achieves **9/10**, with the single failure corresponding to two bonds sharing an endpoint whose wavefronts cancel (the BLUE-BLUE-RED interference pattern discussed in Section 10).

7 Tensor Network Simulation as Quantum Circuit Execution

The Trotter simulation underlying our method is not merely an approximation of a quantum circuit; it *is* the quantum circuit, represented in the Pauli operator basis rather than the amplitude basis. The identification is exact:

- Each **lattice site** = one qubit
- Each **bond gate** $\exp(-i\theta(XX + YY + ZZ))$ = one two-qubit gate
- Each **color class** (non-overlapping bonds applied in parallel) = one gate layer
- Each **belief propagation synchronization** after a full Trotter cycle = one barrier (per-color-class BP, as in the Julia L-MPF implementation [3], provides finer sub-cycle resolution matching the quantum circuit barrier structure)
- Each **PAULI_NL output** = a mid-circuit Pauli weight measurement

This correspondence is not a coincidence: both the graph coloring (determining which bonds share qubits) and the quantum circuit parallelism constraint (which gates can be applied simultaneously) derive from the same source: the lattice topology. The color class ordering is the gate layer ordering.

Two timescales. This identification reveals two distinct timescales:

Physical time t: integer Trotter cycles, $t = n \cdot \varepsilon$. The observables $C(t)$ and $n(v, t)$ at cycle granularity are physical — independent of color class ordering to $O(\varepsilon^2)$ and convergent to true quantum dynamics.

Circuit time t_{color}: sub-cycle, per-color-class granularity. This is a property of the circuit implementation, not the Hamiltonian. Two Trotter implementations of the same Hamiltonian with different color orderings agree on $C(t)$ to first order in ε but differ by $O(\varepsilon^2)$ Trotter errors that accumulate over time; their $n(v, t_{\text{color}})$ tomography fingerprints differ already at the first sub-cycle. The BP synchronization granularity determines which timescale is accessible: per-cycle BP (as used here) loses sub-step information; per-color BP (Julia L-MPF [3]) respects the causal barrier structure of the quantum circuit.

Bond perturbation as gate miscalibration. A perturbed bond (coupling $J \neq 1$) is a *miscalibrated gate*: the two-qubit rotation angle $\theta = J \cdot \varepsilon$ differs from the nominal value. The Pauli transfer matrix for $\exp(-i\theta(XX+YY+ZZ))$ has diagonal entries $\cos(\theta)$, which for any $J \neq 1$ gives $\cos(J\varepsilon) \neq \cos(\varepsilon)$. Since $|\cos(\theta)| \leq 1$ for all θ (unitary gate), any coupling deviation attenuates the wave relative to baseline, guaranteeing $n_{\text{pert}}(v^*) < n_{\text{base}}(v^*)$ at the shadow endpoint v^*, so $\mathcal{L}(v^*) > 0$ (red) regardless of the sign of $\delta J = J - 1$. The detection method is robust: the red site is always the shadow endpoint, never inverted.

This frames our tomography as *quantum ATPG* (Automatic Test Pattern Generation) [4], the classical circuit testing methodology generalized to quantum gates. In classical ATPG: fault activation (excite the stuck-at gate), path sensitization (find an observable path from fault to output), backward implication (infer the faulty gate). Here: perturbation (change J), wavefront propagation (operator spreading from center to shadow site), and cross-entropy inversion (red site + center position $\rightarrow$ bond identity).

8 The Seismic Approach

Running all p centers simultaneously (seismic initialization $O(0) = \sum_k Z_{c_k}$) produces the same ensemble field in a single simulation. At each site v, the first-arrival signal is dominated by the nearest center, automatically selecting the strongest directional component. The sign flip at the broken bond endpoints emerges without any explicit path tracing or center-by-center analysis.

The seismic approach reduces $2p$ simulations (p baselines + p perturbed) to 2 simulations at the cost of mixing center contributions. For the purpose of bond detection the mixing is benign: the sign flip at the broken bond is preserved since the nearest center to each endpoint dominates at t_{first}.

9 Discussion

Connection to hardware. The observable $n(v,t)$ is accessible on real quantum hardware via local Pauli measurements after ZNE gate-folding circuits [1]. The cross-entropy $\mathcal{L}(v)$ requires only the ratio of two measurements — no absolute calibration. The p-center placement is determined by the graph topology alone, independent of the Hamiltonian.

Causal structure. The method exploits the Lieb-Robinson bound: information cannot propagate faster than v_{hop} hops/cycle. A broken bond creates a causal anomaly: the wavefront on one side cannot inform the other. This is detectable without knowing v_{hop} explicitly; the first-arrival time $t_{\text{first}}(v)$ is observed directly.

Complexity and timing. The two methods have different computational costs on the 7×7 lattice ($\chi = 50$, 5 cycles, AMD MI100 32 GB GPU):

Method	Wall-clock	VRAM	Notes
Seismic	272 s	11%	2 GPUs parallel, no v_{hop}
Aperture (no v_{hop})	300 s	11%	3 GPUs, 2× sequential
Aperture ($v_{\text{hop}} = 0.5$)	**191 s**	**5–7%**	3 GPUs, 2× sequential

Table 2: Wall-clock timing comparison on rect 7×7, bond 12, $\chi = 50$, 5 cycles. Both methods produce ✓ CORRECT results. $v_{\text{hop}} = 0.5$ is set by topology ($\text{d}_{\max} = 2$ coverage), not by knowledge of the fault location.

Aperture synthesis with lightcone allocation ($v_{\text{hop}} = 0.5$) is 30% faster than seismic and uses half the VRAM per GPU. The speedup comes from two effects: (1) single-center simulations have smaller active lightcones (χ grows slower), and (2) far centers contribute negligible computation (center 6: 3.5 s, center 34: 9.5 s) while contributing zero to the ensemble, self-filtering at no cost. The bottleneck is the GPU pair running the two nearest centers (25 and 10, ∼20 s each sequential).

Aperture also enables a *coarse-to-fine* strategy unavailable to seismic: run a fast aperture sweep (small χ, $v_{\text{hop}} = 0.5$) to identify candidate bonds, then re-run only the adjacent centers with larger χ for high-accuracy confirmation; each targeted run remains memory-bounded by the lightcone allocation.

Grid-size independence. The most significant property of aperture synthesis with lightcone allocation is that the cost per center is *independent of grid size*. With $v_{\text{hop}} = 0.5$ and n_{steps} cycles, each center sees a local ball of radius $r = n_{\text{steps}} \times v_{\text{hop}} \approx 2.5$ hops, containing ∼13 sites on a 7×7 lattice. The same ball contains the same ∼13 sites on a 100×100 lattice. The computation per center scales as $O(r^2\chi^3)$, where r is the BFS radius, independent of N_{sites}.

Total aperture cost:

$$T_{\text{aperture}} = p \times T_{\text{local}}(\chi, r) \tag{9}$$

where p grows with the number of bonds (more centers needed for coverage) but each center cost T_{local} is constant. For a single bond perturbation, most centers contribute zero signal; their $\mathcal{L}_k(v) \approx 0$ everywhere. Their wall-clock is short (center 6: 3.5 s) and their ensemble contribution is zero. They are neither harmful nor expensive.

Seismic pays $O(N_{\text{sites}}\chi^3)$ per simulation regardless. As the lattice grows, aperture synthesis becomes increasingly favored over seismic. The crossover occurs when $p \times T_{\text{local}} > T_{\text{seismic}}$, which for fixed r and χ requires $p \gtrsim N_{\text{sites}}/r^2$ centers — a condition that is rarely satisfied in practice since the geodesic bond cover grows much slower than the grid area.

Interference signatures. When two adjacent bonds are perturbed simultaneously (bonds 12 and 13, sharing endpoint site 16), the cross-entropy field shows a three-site pattern at 3 cycles: site 15 (red, $\mathcal{L} = +0.55$), site 16 (white, $\mathcal{L} \approx 0$), site 17 (blue). The shared endpoint is the *interference node*: the red signal from bond 12 and the blue signal from bond 13 cancel destructively, exactly as in thin-film interference. The white fringe is not a detection failure; it is the interference pattern. The number of white sites between the red and blue boundaries counts the number of broken bonds in the wall.

This extends the signal vocabulary:

- Single broken bond: **red/blue** (two sites, sign flip)
- Two adjacent broken bonds: **red/white/blue** (three sites, interference node)
- n adjacent broken bonds: **red**, $n-1$ **whites**, **blue**

The current algorithm detects pairs; detecting walls requires a triplet scorer $S_{\text{wall}}(b_1, b_2) = \mathcal{L}(v_a) \times \mathcal{L}(v_c)$ where v_b is the shared endpoint (white site) and v_a, v_c are the outer endpoints.

Adaptive seismic-to-aperture strategy. The seismic and aperture methods are naturally complementary and suggest a two-phase adaptive protocol. In **Phase 1** (seismic), all p centers fire simultaneously: one quantum experiment, minimum cost, producing a map of RED sites that identifies the region of interest. In **Phase 2** (aperture), the RED sites from Phase 1 guide the selection of targeted observation centers geometrically positioned to discriminate between candidate bonds adjacent to those sites. Only $k \ll p$ centers need to be run, each independently.

This mirrors standard practice in other imaging domains: a wide-field scout scan (seismic) followed by targeted high-resolution acquisition (aperture). On quantum hardware, Phase 1 maps directly to a single quantum experiment; Phase 2 selects the minimum number of additional experiments to confirm the bond identity.

The seismic phase is also the most natural starting point on hardware: it requires only one circuit (all centers in a single initialization), produces a rough hypothesis, and costs two experiments (baseline + perturbed). The aperture phase then tests the hypothesis with directional projections. Perturbation signals that survive across independent aperture projections with geometry-consistent timing are physical; chi-truncation artifacts and decoherence errors are uncorrelated across independent runs and average out.

Wavefront triangulation across scans. When multiple independent scans are available, the temporal order in which each scan detects the broken bond carries geometric information beyond the sign flip alone. This enables a triangulation strategy analogous to GPS: each scan provides a time-of-arrival constraint that localizes the bond.

Formally, scan s with nearest center c_s detects the broken bond at cycle t_s^*, the first cycle where $|\mathcal{L}_k(v)| > \varepsilon_{\text{th}}$ for some bond endpoint v. This constrains the bond to lie within $d_s = t_s^* \times v_{\text{hop}}$ hops of c_s: a ball $B(c_s, d_s)$ on the graph. Across M scans, the bond lies in the intersection:

$$b^* \in \bigcap_{s=1}^{M} \mathcal{B}(c_s,\, t_s^* \times v_{\text{hop}}) \tag{10}$$

where $\mathcal{B}(c, r)$ is the set of bonds within crossing distance r of center c.

The sign of the earliest detection (which endpoint is RED vs BLUE) further constrains the direction: the center c_s is on the BLUE (wall-face) side of the broken bond. Subsequent scans confirm or deny: a scan detecting the *opposite* sign at the same endpoints confirms (center is on the other side of the same bond); a scan detecting different endpoints shifts the candidate set. Each confirmation tightens the intersection; each denial eliminates candidates.

In practice on the 7×7 lattice with 3 scans (corners+center, quadrants, mid-edges): scan 2 detects bond 9 (11↔12) at cycle 1 (center 13, 1 hop away), scan 3 confirms with opposite sign at cycle 2 (center 4, 2 hops away), scan 1 provides weak confirmation at cycle 2 from the far corners. The intersection of the three balls gives a small candidate set containing the true bond.

BFS-guided fault detection algorithm. The detection method reduces to a graph-theoretic algorithm that operates entirely on the observable $n(v, t)$, without reference to the Hamiltonian, coupling values, or simulation parameters. We state it precisely.

Let $G = (V, E)$ be the coupling graph, $\{c_s\}$ the observation centers, and ε_{th} the detection threshold.

1. **BFS frontier timing.** For each center c_s and each site $v \in V$, define the *baseline arrival time* $t^*_{\text{base}}(v) = \min\{t : n_{\text{base}}(v, t) > \varepsilon_{\text{th}}\}$ and *perturbed arrival time* $t^*_{\text{pert}}(v) = \min\{t : n_{\text{pert}}(v, t) > \varepsilon_{\text{th}}\}$. These are the BFS frontier crossing times: the actual wavefront arrival, not a predicted hop distance. Note that $t^*_{\text{base}}(v)$ and $t^*_{\text{pert}}(v)$ may differ if a broken bond on the path to v delays the perturbed wave.

2. **Sub-cycle causal ordering.** Within the same BFS cycle, the Trotter color class $c(e) \in \{0, 1, 2, 3\}$ of each bond e determines the firing order. Assign each site v the pair $(h(v), c(e_v))$ where $h(v)$ is its BFS hop depth and $c(e_v)$ is the color class of the bond used to reach it. The total order (h, c) gives exact sub-cycle causality: $(h_u, c_u) < (h_v, c_v)$ means site u is strictly upstream of v.

3. **Bond transfer and anomaly detection.** Consider a bond (u, v) where u is the source (center-side) and v is the destination (shadow-side). The *bond transfer coefficient* is:

$$T = \frac{n(v,\, t^*_{\text{base}}(v))}{n(u,\, t^*_{\text{base}}(u))} \tag{11}$$

When $n_{\text{base}}(u, t^*_{\text{base}}(u)) \gg \varepsilon_{\text{th}}$ (source well established), the threshold condition ensures both waves have the same source amplitude: $n_{\text{base}}(u) \approx n_{\text{pert}}(u)$ at the BFS frontier, since the perturbation is downstream of u. In this regime:

$$\mathcal{L}(v,\, t^*_{\text{base}}(v)) = -\ln \frac{n_{\text{pert}}(v)}{n_{\text{base}}(v)} \approx \ln \frac{T_{\text{base}}}{T_{\text{pert}}} \tag{12}$$

so $\mathcal{L}$ directly measures the *relative change in bond transfer coefficient*. A broken bond ($J \neq 1$) reduces the transfer $T_{\text{pert}} < T_{\text{base}}$, giving $\mathcal{L}(v) > 0$ (RED shadow). The wall face u shows

$\mathcal{L}(u) < 0$ (BLUE accumulation) from the same effect in reverse. A delay $t^*_{\text{pert}}(v) > t^*_{\text{base}}(v)$ is additional evidence that the transfer was slowed.

4. **Bond identification.** For each anomalous site v, examine its neighbors u in G. The broken bond is (u, v) where: (a) $\mathcal{L}(u)$ and $\mathcal{L}(v)$ have opposite signs (sign flip), and (b) $(h_u, c_u) < (h_v, c_v)$ (causal order: u is center-side, v is shadow). Score each candidate bond by $|\mathcal{L}(u) - \mathcal{L}(v)| \cdot \delta/t^*$ where $\delta = 2$ if delayed, 1 otherwise, and $t^* = \min(t^*_{\text{base}}(u), t^*_{\text{base}}(v))$.

5. **Multi-scan aggregation.** Repeat steps 1–4 for each scan independently (no Voronoi spatial restriction). Rank all candidate bonds by total score across scans. Declare the top-k bonds as broken, where k is the number of expected faults.

Pattern vocabulary. The algorithm naturally identifies distinct interference signatures depending on the relative geometry of broken bonds and centers:

- **RED−BLUE**: single broken bond, center on one side: clean sign flip
- **BLUE−WHITE−BLUE**: bond sandwiched between two equidistant centers (collinear): white site is the interference node; both bonds identifiable
- **BLUE−BLUE−RED**: two orthogonal broken bonds at a shared endpoint — shared site accumulates from two directions; outer RED identifies one bond, adjacent BLUEs identify the second

Each pattern is a direct consequence of the BFS frontier geometry and the Trotter color class ordering; no quantum-specific reasoning required.

Experimental results. On a 7×7 lattice with 3 scans (13 centers, 3 GPUs parallel, $d_{\max} = 2$), testing 10 random trials with 2 simultaneous broken bonds ($\delta J = -0.20$): the BFS algorithm with sub-cycle color-class ordering finds **both bonds correctly in 9/10 trials** (90%), with 1 partial case and 0 complete failures. The single partial failure (trial 5) corresponds to two orthogonal bonds sharing an endpoint where both centers' waves are simultaneously blocked: the BLUE−BLUE−RED pattern, representing the hardest geometric configuration.

Failure taxonomy. We characterize five failure modes of increasing complexity:

1. **Single bond** (demonstrated): clean sign flip, 82× margin
2. **Multiple random bonds**: shadows may overlap, L field becomes ambiguous
3. **Collinear bonds**: line of failures looks like a thick wall: bond resolution fails
4. **Hexagonal cluster**: faulty qubit touches 2-3 bonds: RED cluster with BLUE halo
5. **Graduated failure** (δJ smooth variation): dimmer switch; sensitivity limit

Cases 2-5 represent the primary open experimental questions.

Limitations. The current experiment uses the identity PTM, a maximally disruptive perturbation that is deliberately clean. Cases 2-5 above are designed to stress-test the method: case 2 tests whether overlapping shadows can be resolved, case 3 tests spatial resolution limits, case 4 tests cluster detection on irregular topologies, and case 5 tests the sensitivity threshold for realistic coupling deviations.

10 Multi-Bond Detection and Iterative Calibration

Multi-bond detection. We extend the seismic BFS algorithm to detect k simultaneously broken bonds by declaring the top-k candidates from the bond score ranking. Table 3 summarizes the detection rate across 10 single-bond trials (two datasets), 10 two-bond trials, 5 four-bond trials, and 2 eight-bond trials. All experiments use $\delta J = -0.20$, $J = 0.80$.

k bonds	Trials	Bonds found	Per-bond rate
1	20	20/20	100%
2	10	19/20	95%
4	5	18/20	90%
8	2	15/16	94%

Table 3: Per-bond detection rate (single seismic iteration, top-k declaration, $\delta J = -0.20$, 7×7 lattice, $\chi = 50$, 5 cycles). The per-bond detection rate remains above 90% even at 8 simultaneous faults.

The per-bond detection rate remains above 90% even at 8 simultaneous faults. Systematic failures are geometrically localized: bond 8 (sites 10↔11) and bond 36 (sites 43↔44) consistently score below the top-k threshold due to weak seismic illumination from the three scan center placements. These are not random failures; they reflect the coverage geometry of the 3-scan plan.

Iterative calibration loop. The multi-bond results motivate a natural operating protocol:

1. Run seismic MAWGM → declare top-k broken bonds (>90% correct per bond).
2. Recalibrate the declared bonds on hardware.
3. Re-run MAWGM. Fixed bonds disappear from the ranking; remaining faults surface with cleaner signal.
4. Repeat until $n_{\text{pert}}(v,t) \approx n_{\text{base}}(v,t)$ everywhere, with no residual signal.

This iterative approach reduces every multi-bond problem to a sequence of single-bond problems. At 8 simultaneous faults, 7 are found in iteration 1; fixing those removes their interference from the wavefield and the surviving fault becomes easily detectable in iteration 2 (single-bond, 100% rate).

The iterative loop also justifies the design choices: short cycles (3–5) are sufficient because only the next bond to fix needs to be ranked, not all faults simultaneously; a moderate threshold $\varepsilon_{\text{th}} = 10^{-3}$ maximizes BFS coverage so the broken bond is never in a blind spot; and 3 seismic placements cover all 84 bonds with no geometric dead zones except the small set of bonds with weak illumination from all three centers. The stopping criterion (wavefront convergence) is physical and requires no additional measurement overhead beyond the standard MAWGM protocol.

11 Measurability and Detection Thresholds

The detection method requires $n(v,t)$ to be measurable, both from the simulation fidelity standpoint and from the hardware shot budget. Two independent conditions must hold simultaneously.

Condition 1: Statistical measurability (hardware shots). On real hardware, $n(v,t)$ is estimated from N measurement shots as $\hat{n}(v,t) = k/N$ where $k \sim \text{Binomial}(N, n(v,t))$. The shot noise floor is $\sigma(n) \approx \sqrt{n/N}$. For reliable detection of a signal $\delta n = n_{\text{pert}} - n_{\text{base}}$, we require $|\delta n| \gg \sigma(n)$, which gives a minimum measurable n:

$$n_{\min}^{\text{shot}} \approx \frac{1}{\sqrt{N}} \tag{13}$$

N shots	$n_{\min}^{\text{shot}}$	Status
10^3	3×10^{-2}	only strong wavefront
10^4	1×10^{-2}	first-arrival signal
10^5	3×10^{-3}	approaching threshold
10^6	1×10^{-3}	matches simulation floor

Table 4: Minimum measurable $n(v,t)$ as a function of shot budget N. At $N = 10^6$ the statistical floor matches the simulation cutoff.

Condition 2: Simulation fidelity (TN cutoff). In the tensor network simulation, bond singular values below the truncation cutoff δ_{cut} are set to zero. Any $n(v,t)$ arising from these truncated components is an artifact of the TN approximation, not physical operator spreading. The simulation cannot faithfully represent values below δ_{cut}^k, where k is the number of bonds on the shortest path from center to site v. In practice, the effective simulation floor is $\approx \delta_{\text{cut}} = 10^{-3}$.

Furthermore, the belief propagation used to compute messages is not strictly causal: messages propagate globally in each sweep, creating tiny non-zero $n(v,t) \sim 10^{-4}$ at causally disconnected sites (beyond the Lieb-Robinson cone). These BP leakage values are below δ_{cut} and physically unmeasurable on hardware.

The two conditions converge. Setting $\delta_{\text{cut}} = 10^{-3}$ (our default), Condition 2 requires $n_{\min}^{\text{sim}} = 10^{-3}$. From Table 4, Condition 1 requires $N \geq 10^6$ shots to match this floor. The detection threshold $\varepsilon_{\text{th}} = 10^{-3}$ used throughout this work satisfies both conditions simultaneously at $N = 10^6$.

$$\varepsilon_{\text{th}} \geq \max\left(\delta_{\text{cut}},\ \frac{1}{\sqrt{N}}\right) \tag{14}$$

For $N < 10^6$, the hardware floor dominates and a larger ε_{th} is required. For tighter cutoffs $\delta_{\text{cut}} < 10^{-3}$, the simulation floor drops below the hardware floor, enabling more cycles before saturation at the cost of longer simulation time.

Minimum detectable coupling deviation. The signal strength $\delta n(v,t) \approx n_{\text{base}}(v,t) \cdot \varepsilon \cdot \sin(\varepsilon) \cdot |\delta J|$ gives the minimum detectable coupling deviation:

$$|\delta J|_{\min} \approx \frac{\varepsilon_{\text{th}}}{\varepsilon \cdot \sin(\varepsilon) \cdot \sqrt{N \cdot n_{\text{base}}(v,t)}} \tag{15}$$

For $\varepsilon = 0.10$, $N = 10^6$, $n_{\text{base}} \approx 10^{-2}$ (genuine wavefront at first arrival): $|\delta J|_{\min} \approx 1/(0.10 \times 0.0998 \times 316) \approx 0.32$. This improves as $1/\sqrt{N}$ with more shots. For $N = 10^7$: $|\delta J|_{\min} \approx 0.10$.

12 Conclusion

We have demonstrated that the operator lightcone wavemap enables model-free Hamiltonian tomography via a simple signed cross-entropy observable. The broken bond is identified by its causal signature — a sign flip in $\mathcal{L}(v)$ at its two endpoints, with an 82-fold margin using an ensemble of 6 observation centers on a 7×7 lattice. The seismic multi-center initialization achieves the same result in a single simulation.

Extending to realistic coupling degradation ($\delta J = -0.20$, $J = 0.80$) and multiple simultaneous faults, the integrated log-ratio $\Lambda(v) = \ln(\mathcal{B}(v)/\mathcal{N}(v))$ achieves a per-bond detection rate above 90% for up to 8 simultaneous faults in a single seismic pass, using only 6 simulations (3 baseline + 3 perturbed). The iterative calibration loop (detect, fix, re-run) reduces the multi-bond problem to a sequence of single-bond problems, each solved at 100% accuracy.

The method requires no model assumptions and no knowledge of the perturbation type or magnitude. It is a natural sequitur to the Lindblad MPF framework: both use the same observable $n(v, t)$ and the same circuits, but in opposite inference directions: MPF infers noise from a known Hamiltonian, while MAWGM infers coupling geometry from a known noise baseline.

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A Aperture Synthesis with Voronoi Center Placement

An alternative to the seismic approach runs each center independently, in direct analogy with aperture synthesis in radio interferometry. Each center k at a different graph position plays the role of one antenna at a different baseline; $\mathcal{L}_k(v)$ encodes which side of the broken bond site v is on (sign) and the signal strength (magnitude).

Centers are placed with the constraint that no two share the same row or column in the lattice. This guarantees every bond has a strictly unique nearest center (no Voronoi ties): the perpendicular bisector between any two centers never passes through a bond, so one center always dominates the signal at every bond. For $p = 7$ on the 7×7 lattice (one center per row, one per column, a permutation placement), exhaustive search over permutations minimizing $\max_b T_b$ gives centers $\{3, 13, 15, 25, 35, 37, 47\}$ with zero ties across all 84 bonds, maximum $T_b = 5$ cycles, and balanced Voronoi regions of 10–16 bonds each.

Ensemble summation $\mathcal{L}_{\text{ens}}(v) = \sum_k \mathcal{L}_k(v)$ fails for bonds sandwiched between centers on opposite sides, where RED and BLUE contributions partially cancel. A *winner-takes-all* rule avoids

this: for each center k compute the signed contrast across every bond,

$$C_k(b) = \left|\mathcal{L}_k^{\text{signed}}(v_a) - \mathcal{L}_k^{\text{signed}}(v_b)\right| \tag{16}$$

and declare the globally winning center-bond pair $(k^*, b^*) = \arg\max_{k,b} C_k(b)$. The winning center is necessarily on the BLUE side of the broken bond; each center is a fully independent observer and the one with the strongest signal wins. On 10 trials ($\delta J = -0.20$), the adaptive seismic-then-aperture protocol achieves **10/10** correct identifications: 7 bonds identified by seismic alone (2 simulations), 3 by aperture (14 simulations), at an average cost of $2 + 0.3 \times 14 \approx 6$ simulations per bond.

The seismic 3-scan approach in Section 6 supersedes this method for the multi-bond case: it achieves the same single-bond accuracy at lower cost, and extends naturally to multiple simultaneous faults without requiring the no-shared-row-or-column placement constraint.